

HAMZAVERSE ENGINEERING TOOLS

The ChemE Fluid & Heat Formula Bible

Every formula you need for Fluid Mechanics and Heat Transfer coursework — explained, worked, and ready to use. Built by a Chemical Engineering student, for Chemical Engineering students.

Reynolds Number

Friction Factor

Head Loss

Pump Power

NPSH

LMTD

Heat Exchanger Area

How To Use This Guide

A quick-reference companion for exams, assignments, and lab reports — not a textbook replacement.

Each topic gets one page: the formula, what every symbol means, when to use it, and one fully worked example with the units carried through. Numbers are checked step-by-step, so you can trust them in a lab report.

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A NOTE ON UNITS

Every formula here uses **SI units** (kg, m, s, Pa, W) unless stated otherwise. Mixing unit systems is the #1 cause of wrong answers — see page 12.

Continuity & Bernoulli

Every fluid mechanics problem below builds on these two equations.

Continuity Equation (mass conservation)

$$A_1 V_1 = A_2 V_2 \quad | \quad Q = A \cdot V$$

A = cross-sectional area (m²), V = velocity (m/s), Q = volumetric flow rate (m³/s)

What flows in must flow out. If a pipe narrows, velocity must increase to keep Q constant.

Bernoulli Equation (energy conservation, incompressible flow)

$$P_1/\rho g + V_1^2/2g + z_1 = P_2/\rho g + V_2^2/2g + z_2 + h_L$$

P = pressure (Pa), ρ = density (kg/m³), g = 9.81 m/s², z = elevation (m), h_L = head loss (m)

Each term has units of length ("head"): pressure head, velocity head, elevation head. Adding h_L accounts for real friction — this is what pages 6-8 calculate.

WORKED EXAMPLE

A pipe narrows from D₁ = 100 mm to D₂ = 50 mm. If V₁ = 1 m/s, find V₂.

$$A_1 V_1 = A_2 V_2 \rightarrow V_2 = V_1 (D_1/D_2)^2 = 1 \times (100/50)^2$$

$$V_2 = 4 \text{ m/s}$$

Reynolds Number

Tells you whether flow is smooth (laminar) or chaotic (turbulent) — decides which formula you use next.

$$Re = \rho V D / \mu = V D / \nu$$

ρ = density (kg/m³), V = velocity (m/s), D = pipe diameter (m), μ = dynamic viscosity (Pa·s), ν = kinematic viscosity (m²/s)

Flow Regime (internal pipe flow)

Range	Regime	What it means
$Re < 2,300$	Laminar	Smooth, layered flow — use $f = 64/Re$
$2,300 - 4,000$	Transitional	Unstable, avoid designing in this range
$Re > 4,000$	Turbulent	Chaotic mixing — use Colebrook / Swamee-Jain

Some textbooks use 2,100 as the laminar cutoff instead of 2,300 — check which your course uses.

WORKED EXAMPLE

Water at 20°C ($\rho = 998 \text{ kg/m}^3$, $\mu = 1.002 \times 10^{-3} \text{ Pa}\cdot\text{s}$) flows at $V = 1 \text{ m/s}$ through $D = 0.05 \text{ m}$ pipe.

$$Re = (998 \times 1 \times 0.05) / 1.002 \times 10^{-3}$$

$Re \approx 49,800 \rightarrow \text{Turbulent}$

Friction Factor

Feeds directly into the head loss equation on the next page.

Laminar Flow ($Re < 2,300$)

$$f = 64 / Re$$

Exact solution — no approximation needed.

Turbulent Flow — Swamee-Jain (explicit approximation)

$$f = 0.25 / [\log_{10} (\epsilon / 3.7D + 5.74 / Re^{0.9})]^2$$

ϵ = pipe roughness (m) — see table below. Accurate within ~1% of Colebrook for $Re: 5,000-10^8$, $\epsilon/D: 10^{-1}-0.05$

Typical Roughness (ϵ)

Material	ϵ (mm)
PVC / Glass (smooth)	0.0015
Commercial steel	0.045 – 0.046
Galvanized iron	0.15
Cast iron	0.26
Concrete	0.3 – 3.0

WORKED EXAMPLE

$Re = 99,600$, commercial steel $D = 0.05$ m, $\epsilon = 0.046$ mm $\rightarrow \epsilon/D = 0.00092$

$$f = 0.25 / [\log_{10}(0.00092/3.7 + 5.74/99,600^{0.9})]^2$$

$$f \approx 0.0221$$

Head Loss (Major & Minor)

This is the h_L term in Bernoulli's equation — real friction losses along a pipe run.

Major Loss (pipe friction)

$$h_f = f \cdot (L/D) \cdot (V^2/2g)$$

L = pipe length (m), D = diameter (m), f from page 5

Minor Loss (fittings)

$$h_m = K \cdot (V^2/2g)$$

K = loss coefficient (see table)

$$h_L = h_f + \Sigma h_m$$

Typical K-Values (vary by manufacturer — treat as estimates)

Fitting	K	Fitting	K
Sharp entrance	0.5	Gate valve (open)	0.2
Rounded entrance	0.05	Globe valve (open)	10
Pipe exit	1.0	Swing check valve	2.0
90° smooth elbow	0.3	Tee (line flow)	0.2
90° threaded elbow	1.5	Tee (branch flow)	1.0

WORKED EXAMPLE

Water, $V = 2$ m/s, $D = 0.05$ m, $L = 100$ m, commercial steel ($f = 0.0221$ from pg 5)

$$h_f = 0.0221 \times (100/0.05) \times (2^2 / (2 \times 9.81))$$

$$h_f \approx 9.0 \text{ m}$$

Pump Power & Efficiency

How much power a pump must deliver to move fluid against head H.

Hydraulic (Water) Power

$$P_{\text{hydraulic}} = \rho \cdot g \cdot Q \cdot H$$

Q = flow rate (m³/s), H = total dynamic head (m) — includes elevation + friction losses

Shaft (Brake) Power — what the motor must supply

$$P_{\text{shaft}} = P_{\text{hydraulic}} / \eta$$

η = pump efficiency (as a decimal, e.g. 70% = 0.70). Typical centrifugal pumps: 50-85%.

WORKED EXAMPLE

Q = 10 L/s = 0.010 m³/s, H = 30 m, η = 70%

$$P_{\text{hydraulic}} = 998 \times 9.81 \times 0.010 \times 30 \approx 2,938 \text{ W}$$

$$P_{\text{shaft}} = 2,938 / 0.70$$

$$P_{\text{hydraulic}} \approx 2.94 \text{ kW} \mid P_{\text{shaft}} \approx 4.20 \text{ kW}$$

WATCH OUT

Always convert Q to m³/s before using this formula. L/s and m³/hr are the most common unit traps here.

NPSH — Avoiding Cavitation

Net Positive Suction Head: the safety margin against a pump boiling its own fluid at the impeller inlet.

$$\text{NPSH}_a = (P_{\text{atm}} - P_{\text{vapor}}) / (\rho g) \pm h_s - h_{f,\text{suction}}$$

h_s is + if the liquid source is above the pump (flooded suction), – if below (suction lift). $h_{f,\text{suction}}$ = friction loss in the suction line only.

The Rule

$$\text{NPSH}_a > \text{NPSH}_r \quad (\text{with margin, typically } +0.5\text{--}1 \text{ m})$$

NPSH_r (required) comes from the pump manufacturer's curve — it is not calculated, it's read off a datasheet.

If NPSH_a drops below NPSH_r , the fluid flashes to vapor at the impeller eye, collapses violently, and pits the metal — this is cavitation, and it destroys pumps.

WORKED EXAMPLE

Open tank, water 20°C, source is 2 m above pump ($h_s = +2 \text{ m}$), suction friction loss = 0.5 m.

At 20°C: $P_{\text{atm}}/\rho g \approx 10.33 \text{ m}$, $P_{\text{vapor}}/\rho g \approx 0.24 \text{ m}$

$$\text{NPSH}_a = 10.33 - 0.24 + 2 - 0.5$$

$$\text{NPSH}_a \approx 11.6 \text{ m}$$

LMTD — Heat Exchangers

Log Mean Temperature Difference: the correct "average" ΔT to use when temperature difference changes along the exchanger.

$$\text{LMTD} = (\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2)$$

Counterflow

$$\Delta T_1 = T_{h,in} - T_{c,out}$$

$$\Delta T_2 = T_{h,out} - T_{c,in}$$

Parallel Flow

$$\Delta T_1 = T_{h,in} - T_{c,in}$$

$$\Delta T_2 = T_{h,out} - T_{c,out}$$

WORKED EXAMPLE

Counterflow: Hot fluid $150^\circ\text{C} \rightarrow 90^\circ\text{C}$, Cold fluid $30^\circ\text{C} \rightarrow 70^\circ\text{C}$

$$\Delta T_1 = 150 - 70 = 80^\circ\text{C} \quad \Delta T_2 = 90 - 30 = 60^\circ\text{C}$$

$$\text{LMTD} = (80 - 60) / \ln(80/60)$$

$$\text{LMTD} \approx 69.5^\circ\text{C}$$

WATCH OUT

Counterflow and parallel flow pair up the temperatures differently. Mixing them up is the most common LMTD mistake.

Heat Exchanger Area

Once you have LMTD, this tells you how big the exchanger needs to be.

$$Q = U \cdot A \cdot \text{LMTD}$$

Q = heat duty (W), U = overall heat transfer coefficient ($\text{W}/\text{m}^2\text{K}$), A = heat transfer area (m^2)

$$A = Q / (U \cdot \text{LMTD})$$

Typical U-Values (rough design estimates)

Service	U ($\text{W}/\text{m}^2\text{K}$)
Water – Water	800 – 1,500
Water – Light oil	100 – 350
Water – Gas	10 – 50
Steam condenser (water)	1,500 – 4,000

Always use the value given in your problem statement — this table is only for sanity-checking your answer.

WORKED EXAMPLE

$Q = 500 \text{ kW} = 500,000 \text{ W}$, $U = 500 \text{ W}/\text{m}^2\text{K}$, $\text{LMTD} = 69.5^\circ\text{C}$ (from pg 9)

$A = 500,000 / (500 \times 69.5)$

$A \approx 14.4 \text{ m}^2$

One-Page Formula Summary

Every formula in this guide, in one place — for the last five minutes before an exam.

Topic	Formula
Continuity	$A_1V_1 = A_2V_2$
Bernoulli	$P/\rho g + V^2/2g + z = \text{const.} + h_L$
Reynolds number	$Re = \rho V D / \mu$
Friction factor (laminar)	$f = 64/Re$
Friction factor (turbulent)	$f = 0.25 / [\log_{10}(\epsilon/3.7D + 5.74/Re^{0.9})]^2$
Major head loss	$h_f = f(L/D)(V^2/2g)$
Minor head loss	$h_m = K(V^2/2g)$
Hydraulic power	$P = \rho g Q H$
Shaft power	$P_{\text{shaft}} = P_{\text{hydraulic}} / \eta$
NPSH available	$NPSH_a = (P_{\text{atm}} - P_v) / \rho g \pm h_s - h_f$
LMTD	$(\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2)$
Heat exchanger duty	$Q = U \cdot A \cdot LMTD$

Key Constants

Constant	Value
g (gravity)	9.81 m/s ²
Water density (20°C)	998 kg/m ³
Water viscosity (20°C)	1.002×10^{-3} Pa·s
Standard atmosphere	101,325 Pa $\approx$ 10.33 m water head

Unit Traps & Common Mistakes

Most wrong answers in fluid mechanics come from units, not concepts.

Trap	Fix
Flow rate given in L/s or L/min	Convert to m ³ /s: divide L/s by 1,000
Diameter given in mm or inches	Convert to metres before using in Re, f, h _f
Viscosity given in centipoise (cP)	1 cP = 1×10 ⁻³ Pa·s
Temperature in °C used in ratios (e.g. gas laws)	Convert to Kelvin: K = °C + 273.15
Efficiency given as a percentage	Divide by 100 before use (70% → 0.70)
Roughness (ε) in mm, diameter (D) in m	Convert both to the same unit before ε/D
Using parallel-flow ΔT pairing for a counterflow exchanger	Re-check which stream layout the problem describes

Sanity Checks Before You Submit

CHECK 1

Is the flow regime (laminar/turbulent) consistent with which friction factor formula you used?

CHECK 2

Does your head loss make physical sense? (A few metres per 100 m of pipe is typical — 500 m of head loss signals a unit error.)

CHECK 3

Is $NPSH_a$ actually greater than $NPSH_r$, with a safety margin?

KEEP GOING

Check Your Work Instantly

Every formula in this guide has a free calculator on HamzaVerse Engineering Tools — plug in your numbers, see the flow regime, and get step-by-step output instead of just a final answer.

hamzaverse77-lang.github.io

Built by a Chemical Engineering student. Found an error or want a topic added? Message on Reddit or via the site's contact page.